

Question Bank – Computational Theory – PT 1

- Q. 1. Write Regular Expression for
- Set of all strings over $\{0, 1\}$ that end with 1 has no substring 00
 - Set of all strings over $\{0, 1\}$ with even number of 1's followed by odd number of 0's
 - $L = \{w \mid w \text{ has prefix } bab \text{ and suffix } abb \text{ and } w \text{ is a string over } \{a, b\}\}$
 - All strings starting with b and ending in ba over $\{a, b\}$
- Q. 2. Define Regular Expression
- Q. 3. Describe Finite State Machine
- Q. 4. Give applications of Regular Expressions and Finite Automata
- Q. 5. Explain Chomsky Hierarchy with examples
- Q. 6. Design a DFA over $\Sigma = \{a, b\}$ to accept string which ends with 'bb'
- Q. 7. Design a DFA over $\Sigma = \{a, b, c\}$ to accept string which starts and ends with same letter.
- Q. 8. Design a DFA for $L = \{(ab)^n \mid n \geq 0\}$ where $\Sigma = \{a, b\}$
- Q. 9. Design DFA over $\Sigma = \{0, 1\}$ to accept strings containing 101 or 110
- Q. 10. Design DFA to determine whether ternary number (base 3) is divisible by 5
- Q. 11. Let G be the grammar:
- $$S \rightarrow aB \mid BA$$
- $$A \rightarrow a \mid aS \mid bAA$$
- $$B \rightarrow b \mid bS \mid aBB$$
- Find: i) Left most derivation
ii) Right most derivation
iii) Parse Tree
iv) Derive the given string aabbabba and aababb
(try for both using Leftmost and Rightmost derivation)
- Q. 12. For Given Grammar $G = (\{S, T\}, \{0, 1\}, P, S)$
Where P Consists of,
- $$S \rightarrow T00T$$
- $$T \rightarrow 0T \mid 1T \mid \epsilon$$
- Apply Leftmost and Rightmost Derivation for string 1000111.

Q. 13 For Given Grammar $G = (\{A, B, S\}, \{a, b\}, P, S)$

Where P Consists of,

$$S \rightarrow AB$$

$$B \rightarrow a \mid Sb$$

$$A \rightarrow Aa \mid bB$$

Derive String w using Leftmost and Rightmost Derivation.

Draw Derivation Tree where $w = "baabaab"$.

Q. 14 Consider the grammar

$$S \rightarrow oSo \mid 1S1 \mid SS \mid \epsilon$$

Given the string 0101101110, find a leftmost derivation and a rightmost derivation with corresponding parse trees.

Q. 15 Construct CFG for the following:

i. $L(G) = \{a^{2n} b^{2n} \mid n \geq 1\}$

ii. $L(G) = \{a^n b a^m \mid n, m \geq 1\}$

iii. For the language over $\Sigma = \{a, b\}$ containing at least one occurrence of 'aa'

iv. $L(G) = \{a^n b^n c^m d^{2m} \mid m, n \geq 1\}$

Q. 16 Convert the following Right-Linear Grammar to its equivalent Left-Linear Grammar

$$S \rightarrow b B$$

$$B \rightarrow b C$$

$$B \rightarrow a B$$

$$B \rightarrow b$$

$$C \rightarrow a$$

Q. 17 Convert the following Left-Linear Grammar to its equivalent Right-Linear Grammar

$$S \rightarrow B 1 \mid A o \mid C o$$

$$B \rightarrow B 1 \mid 1$$

$$C \rightarrow A o$$

$$A \rightarrow C o \mid A 1 \mid B 1 \mid o$$

Q. 18 Construct a Finite Automata recognizing $L(G)$ where G is a grammar given by

$$S \rightarrow a S \mid b A \mid b$$

$$A \rightarrow a A \mid b S \mid a$$

Q. 19 Construct regular grammar for given DFA which is described as

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$\text{Where } Q = \{ q_1, q_2, q_3, q_4 \}$$

$$\Sigma = \{ a, b \}$$

$$q_0 = q_1$$

$$F = \{ q_4 \}$$

Transition Function δ can be written as

$$\delta (q_1 , b) = q_2 \qquad \delta (q_1 , a) = q_3$$

$$\delta (q_2 , a) = q_3 \qquad \delta (q_2 , b) = q_4$$

$$\delta (q_3 , a) = q_4 \qquad \delta (q_3 , b) = q_2$$

$$\delta (q_4 , a) = q_4 \qquad \delta (q_4 , b) = q_4$$